

39) Express each of these statements using predicates and quantifiers.

a) A passenger on an airline qualifies as an elite flyer if the passenger flies more than 25,000 miles in a year or takes more than 25 flights during that year.

Domain: Set of all passengers on an airline.

$EF(x)$: x is an elite flyer

$M(x, y, z)$: x flies more than y miles in a year z

$F(x, y, z)$: x takes more than y flights in the year z

$$\forall x \exists z (M(x, 25000, z) \vee F(x, 25, z) \Rightarrow EF(x))$$

b) A man qualifies for the marathon if his best previous time is less than 3 hours and a woman qualifies for the marathon if her best previous time is less than 3.5 hours.

Domain: Set of all people.

$M(x)$: x is a man $W(x)$: x is a woman

$T(x, y)$: x 's previous best time is less than y hours.

$$\forall x ((M(x) \wedge T(x, 3)) \Rightarrow Ma(x)) \wedge (W(x) \wedge T(x, 3.5) \Rightarrow Ma(x))$$

c) A student must take at least 60 hours course, or at least 45 hours course and write a master's thesis, and receive a grade no lower than a B in all required courses, to receive a master's degree.

Domain: Set of all students.

$C(x, y)$: x takes at least y course hours $M(x)$: x has written a master's thesis

$G(x, y)$: x has received a grade no lower than y in all required courses

$Ma(x)$: x has received a master's degree.

$$\forall x (Ma(x) \Rightarrow (C(x, 60) \vee (C(x, 45) \wedge M(x) \wedge G(x, B))))$$

d) There is a student who has taken more than 21 credit hours in a semester and received all A's.

Domain: Set of all students

$C(x, y)$: x has taken more than y credit hours in a semester.